SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Data Flow Diagram

Date	06 July 2026
Project Name	**Rising Waters**
Maximum Marks	2 Marks

Data Flow Diagram

Data Flow Explanation Table – Rising Waters (Flood Prediction System)

Element (Type)	ID / Name	Description	Input	Output	Data Store (Used)	Explanation
External Entity	User	The user interacts with the system by providing weather related parameters.	-	• Weather Parameters (Rainfall, Temp., Humidity, Cloud Cover, Wind Speed, Seasonal Rainfall, etc.)	-	The user enters required weather details on the web application to get flood prediction.
Process (1)	Input Validation (Flask)	Validates the inputs received from the user to ensure they are in correct format and within acceptable range.	Weather Parameters from User	Validated Data	-	The Flask application checks for missing values, data type and valid range before processing further.
Process (2)	Data Preprocessing (StandardScaler Transformation)	Transforms the validated data using StandardScaler to match the model's training conditions.	Validated Data	Processed Data (Scaled Features)	D3 – Scaler File (transform.save)	The scaler file is loaded and applied to the input data to normalize the values.
Process (3)	ML Prediction Model (XGBoost)	Uses the trained XGBoost model to predict whether flood will occur or not.	• Processed Data from Process (2) • Raw Data for training	Prediction (Flood / No Flood)	• D1 – Flood Dataset (Excel File) • D2 – Trained Model (flood_model.pkl)	The preprocessed data is passed to the trained model which generates the flood prediction.
Process (4)	Generate Result	Generates user friendly result and prepares the response to be displayed on the web page.	Prediction from Process (3)	Display Result	-	Based on the prediction, the appropriate result page (chance.html / no_chance.html) is selected.
External Entity	Result to User (chance.html / no_chance.html)	Displays the final prediction result and information to the user.	Display Result from Process (4)	Result shown to User	-	The user views the prediction result on the web page.
→ Data Flow	Data Flow (Arrows)	Shows the direction of data movement between entities, processes and data stores.	-	-	-	Arrows indicate how data flows from one component to another in the system.

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